Corridor Reuse Outperforms Bathymetry in Submarine Cable Route Prediction: A Controlled Evaluation on 412 As-Laid Routes

Pavitra Sharma

***Abstract*—Submarine cable route planning tools—academic, commercial and patented—optimise a least-cost path over seabed terrain, on the premise that real cable routes are largely the outcome of such an optimisation. That premise is widely assumed and, to our knowledge, never measured against cables that were actually laid. We assemble a corpus of 412 as-laid routes (206,175 km) republished by seven national hydrographic offices, of which 355 have matched bathymetry, and evaluate against grids at 460 m and 1.85 km. Three controls carry the evaluation: a *placebo-displaced* baseline that establishes what each instrument reads when no preference exists, a *discretisation* control that measures what a grid search costs before it is compared with an analytic baseline, and a *resolution gate* that quantifies the sensitivity lost when the grid is coarsened. Terrain preference is real but small—slope -1.65 percentile points against the placebo—and although adding it to a least-cost search cuts median prediction error from 9.7 km to 7.1 km, *fitting* the cost weights does not beat guessing them. A competing signal is markedly stronger. Cables sit 29% closer to other operators' cables than displaced controls do, on 74% of routes, after excluding same-agency data, the landfall approaches and every cable sharing a named landing point. Built into the same search, this corridor term is the only method tested that beats a great circle, and it *strengthens* with route length, winning on 15/15 routes above 2,000 km ($p = 3\times10^{-4}$). Its operating range is predictable: a per-route rule (Spearman $\rho = 0.656$, $n = 348$) states in advance when it will fail. A third candidate driver, fishing effort, shows no signal once landfalls are removed. Terrain-driven route optimisation models the weaker of two available signals while ignoring the stronger one, which requires no bathymetry at all.**

***Index Terms*—Submarine cables, route planning, least-cost path, model validation, spatial statistics, placebo control, bathymetry, marine spatial planning.**

I. Introduction

SUBMARINE telecommunications cables carry almost all intercontinental data traffic, and where one is laid determines its cost, its permitting burden and its exposure to the hazards that will eventually break it. The dominant computational approach to choosing that path is the least-cost path: rasterise the seabed, assign each cell a cost from depth, slope, roughness and a hazard term, and search for the cheapest route between two landing points [1]–[4]. Commercial packages implement the same idea [5], and a granted patent extends it by *learning* the cost surface from bathymetry around existing routes [6].

The author is with [Affiliation], [City], [Country] (e-mail: pavitrasharma2306@gmail.com).

Analysis code and the full reproduction pipeline are available at https://github.com/ixpavi/subsea-siting.

Every one of these rests on a premise: that the position of a real submarine cable is substantially explained by seabed terrain. The premise is rarely stated and, as far as we can establish, never measured. The most recent comparative study of cable path planning algorithms evaluates Fast Marching, a Dijkstra variant and a great circle against *modelled cost*, describes its own output as "an initial guideline or benchmark rather than the final route", and reports no accuracy metric against any deployed cable [7]. A GIS multi-criteria study applied to real case studies likewise reports no comparison with the cables actually laid [3]. The patent trains a model on existing routes precisely *because* it assumes those routes optimise over bathymetry, and reports no validation of that assumption [6].

This paper supplies the missing measurement, and finds that the premise is weak.

A. Why the measurement is harder than it looks

Three properties of this problem defeat a naive evaluation, and each produced a wrong headline in our own work before it was controlled for.

The null is not where you think it is. Seabed terrain is strongly spatially autocorrelated. Any line drawn on it—cable or not—will score off the nominal midpoint of a local distribution, so a percentile near 50 cannot be read as "no preference" without measuring what a non-cable reads.

The baseline is not on the same footing. A great circle is a smooth analytic curve; a grid search is not. Comparing them attributes the cost of discretisation to the cost function.

The instrument changes when the problem does. Long-haul routing cannot be searched at the resolution regional routing can, so the grid must coarsen, and a null result on a coarser grid is indistinguishable from an instrument too blunt to see the effect.

B. Contributions

1) **A validation of the least-cost-path premise against as-laid routes.** On 412 routes from seven national hydrographic offices, we report prediction error in kilometres against the cable actually laid, for six methods sharing one search, one grid and one endpoint treatment (Section V-D).
2) **Three transferable controls**—placebo displacement, a discretisation penalty, and a resolution gate—each of which changed a headline here (Section IV). We claim

the *transfer and the measured consequence of omitting them*, not the designs; Section II gives their homes in other literatures.

3) **A stronger, bathymetry-free signal.** Corridor reuse survives same-agency exclusion, landfall trimming, shared-landing-point exclusion, and complete replacement of the candidate set, and it strengthens with route length (Sections V-E–V-H).
4) **A rule that predicts its own failure.** Corridor following helps only when the neighbour geometry's error is small relative to the prediction's scale; tested per route it is monotonic across six bins (Section V-I).
5) **A tested and rejected alternative explanation.** Fishing effort, the mechanism behind most cable faults, shows no relationship to cable position once landfalls are removed (Section V-J).

II. Related Work

A. Least-cost-path cable routing

Least-cost path and A* over a rasterised cost surface is textbook, and its application to submarine cables is a small but active literature dominated by one group [1], [2], [4], [7], [8]. Multi-criteria decision analysis over normalised weighted criteria is likewise standard, and has been applied to cable siting with seismic fault avoidance [3].

Nothing in this paper claims a new routing algorithm. Our search is deliberately conventional; it exists so that the premise under test has a concrete instance. These are well-built optimisers whose objective function has not been checked against reality—a gap their authors state themselves [7]—rather than incorrect ones.

B. Learning route cost from observed paths

US 11,031,757 B2 [6] claims generating cable routes from a model trained on bathymetry and existing route data. It is prior art for *learning a cost surface*, which is therefore not claimed here. It is also the strongest available argument that the question matters: a major operator built machinery on the assumption that existing routes encode bathymetric optimisation, without publishing a test of it.

C. What is not claimed as novel

Comparing an observed path against displaced alternatives that share its geometry is the foundational design of step-selection analysis in movement ecology—a matched case–control used-availability design [9]–[11]. The shift-and-rotate null model [12] is closer still: it preserves a predictor's spatial structure while destroying its alignment with the response, which is exactly the logic of displacing a cable sideways. We claim the transfer to *engineered linear infrastructure* and the measured consequence of omitting it, not the design. A known caveat transfers with it: such nulls are sensitive to linear trend [13], and a control displaced along a heading may inherit a depth gradient—which is why Section V-B reports depth separately and finds it moves the other way.

Likewise, reporting the share of admissible weightings under which a recommendation holds is rank acceptability analysis, which has a name and a literature [14], [15].

TABLE I
As-laid route corpus by publishing agency. Sampling fidelity spans a factor of 30, which §V-C shows is confounded with terrain.

Source	Routes	km	vtx/1,000 km	Median seg.
DE BSH-CONTIS	23	2,224	3,369	6 m
UK OGA	2	130	692	327 m
NL Rijkswaterstaat	94	20,398	378	703 m
FR SIGCables	52	23,147	168	2.49 km
FR SHOM	217	156,153	148	2.11 km
MT IOI-MOC	5	845	91	3.07 km
ES CICA	19	3,278	89	6.02 km
TeleGeography	*724*	—	*7.3*	*69 km*

III. Data

A. As-laid routes

The geometry published on cable maps is schematic. Measured directly, a widely used commercial dataset [16] carries 7.3 vertices per 1,000 km, a median segment of 69 km, and a longest single segment of 5,650 km. A 5,650 km straight line records that two places are connected, not where the cable goes; it cannot support any claim about route position.

We therefore use EMODnet Human Activities [17], which republishes cable positions from national hydrographic offices—the data fishermen and marine planners use to avoid them. Telecommunication layers only; power cables have different burial rules and route economics. Screening for routes long enough to have had alternatives and sampled finely enough to show which was taken ($\geq$ 25 km, $\geq$ 50 vtx/1,000 km) gives 412 routes and 206,175 km (Table I).

The corpus is Europe-weighted but not shelf-limited (Fig. 1): the French SHOM layer covers overseas territories, reaching the Pacific, Indian Ocean and Caribbean. Median depth along the corpus is 64 m, but 17% lies below 2,000 m and 4% is abyssal, to a maximum of 5,272 m. Deep-water routing decisions are present.

B. Bathymetry

Two grids are used, and they are not interchangeable.

460 m, downsampled from EMODnet Bathymetry DTM at native 1/16 arc-minute ($\sim$115 m) [18], comfortably finer than the corpus's 2 km median route sampling—the condition for resolving what routes bent around. The downsampling was validated rather than assumed: only 0.1% of output cells exactly equal a source cell, so the service interpolates rather than point-samples; global mean signed error is -1.2 m; the residual is interpolation noise concentrated on steep cells (steep-decile RMS 47 m) against relief of thousands of metres. Of 412 routes, 356 fall inside the product's envelope and one crosses a genuine coverage hole and is excluded rather than interpolated over, leaving 355 routes, which carry every terrain result below.

1.85 km, a uniform 60 cells/degree grid covering the search corridor of every corpus route, back-filled from the NOAA NCEI global DEM mosaic [19] where EMODnet does not reach. Section V-G explains why this grid is unavoidable and Section IV-D measures what it costs.

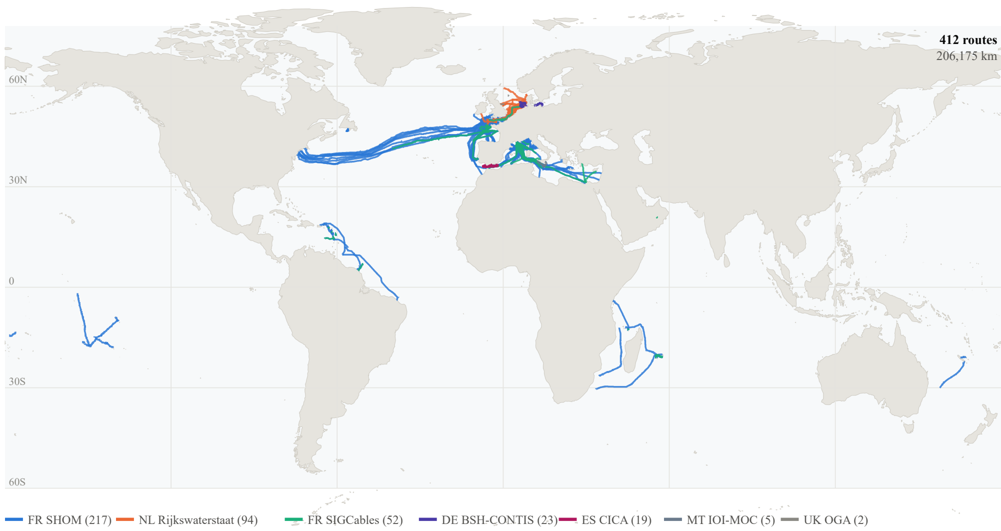

Fig. 1. Geographic coverage of the corpus, coloured by publishing agency. Three quarters of corpus length is northwest Europe and the Mediterranean, and the trans-oceanic reach comes almost entirely from one agency's overseas territories — the concentration Section VI-B reports, shown rather than asserted. It is also why Section V-C's ruggedness gradient is confounded: in this corpus, agency and sea are not separable.

A data defect worth reporting. EMODnet encodes out-of-coverage as exactly 0 rather than as a nodata sentinel: 4.85% of cells across the tiles we built, with 36 tiles entirely zero, among them Barbados and a mid-Atlantic block that are simply outside the DTM. NOAA's exact-zero rate on the same footprint is 0.03%, i.e. genuine sea level. Averaged into a block mean this fabricates a shallow shelf where there is no data; and because depth is read as null for any elevation ≥ 0, such a cell reads as *land*, so a router treats open Atlantic as coastline and confidently goes around it. We discarded 2,203,460 such cells and back-filled them. Once the fill is excluded the two products agree where both report seabed—mean signed difference $+5.0$ m, median $|\text{difference}|$ 18 m—against an apparent $-1{,}550$ m disagreement before it was excluded.

C. Fishing effort

EMODnet Human Activities vessel density, fishing subset, AIS-derived at $\sim$1.7 km native resolution, hours per km^2 per month, 2022 epoch [20]. Global Fishing Watch [21] is the citable global source, but its public grid is $0.1°$ ($\sim$11 km), which cannot resolve a decision on a 1.85 km routing grid. 82.2% of the 10,450 route sample points are scoreable; the remainder lie outside coverage, chiefly Pacific and southern-hemisphere routes.

IV. Method

A. The placebo-displaced control

At each point along a route we take a transect perpendicular to the local heading and ask where the cable sits within it.

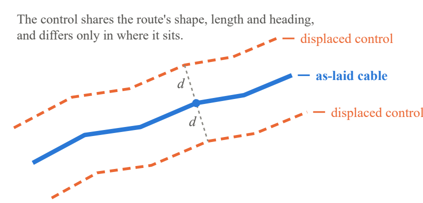

Fig. 2. The placebo-displaced control. The control shares the route's shape, length and heading and differs only in position. Displacement is perpendicular to the *local* heading, so the separation d is constant along a turning route.

The measurement is purely local: no endpoints, no geodesic, no assumption about what the route was for. Under the null the expected percentile is 50. Each route contributes one number (the mean of its samples), because adjacent samples share terrain and pooling them would inflate significance enormously.

The nominal null of 50 is not trustworthy, for the reason given in Section I-A. So the identical measurement is repeated on the same routes displaced sideways by 20 and 50 km. Displacement is perpendicular to the *local* heading rather than a rigid translation, so the separation stays constant along a turning route (Fig. 2). Displaced lines are not cables and cannot express engineering preference, but they share shape, length, heading and terrain—so whatever they score is what the instrument reads when no preference exists.

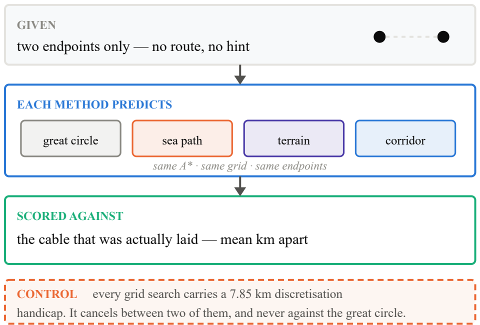

Fig. 3. The evaluation protocol. Every method receives only the two endpoints and is scored on distance from the cable actually laid. All terrain-aware methods share one search, one grid and one endpoint treatment, so a difference between them is attributable to the cost weights rather than to the machinery.

B. Route prediction and its baselines

Each method is given *only* the two endpoints, predicts the route, and is scored by the mean distance from the cable actually laid (Fig. 3). Every terrain-aware method uses the same A* over the same bathymetry, differing only in cost weights, so a difference between them cannot come from the search, the grid, the corridor bound or endpoint handling. Weight fitting uses leave-one-source-out cross-validation, because routes from one agency share a sea and a survey convention; a fold is only run where at least eight routes remain.

C. The discretisation control

The geodesic is analytic. Every other method is a path over an 8-connected grid that can step only in 45° increments and must snap its endpoints to water cells. Those are properties of the *search*, not of terrain.

We therefore measure the zero-weight router—terrain entirely ignored, so over open water it is trying to reproduce the great circle—against the true great circle, on the 51 route pairs whose geodesic never touches land. The median penalty is 7.85 km. That handicap is larger than the 2.63 km gap between the geodesic and the shortest sea path, so "A* is worse than a straight line" measures the grid rather than the bathymetry and cannot support a claim in either direction. Only comparisons between two grid searches, which share the handicap exactly, are sound.

D. The resolution gate

A* searches a corridor around the endpoints' bounding box, whose area grows with the *square* of route length. At 240 cells/degree the median corridor is 478,000 cells for a 40–500 km route and 37.5 million for a 3,000 km one, against a 2,000,000 expansion cap: long-haul at 460 m is not slow, it is impossible.

TABLE II
THE RESOLUTION GATE. THE IDENTICAL PLACEBO MEASUREMENT AT BOTH GRID RESOLUTIONS ON THE SAME 235 ROUTES, REPORTED BEFORE ANY LONG-HAUL RESULT IS QUOTED. SAME SIGN THROUGHOUT; ROUGHLY HALF THE SENSITIVITY SURVIVES ON SLOPE AND RELIEF, ALMOST ALL OF IT ON DEPTH.

Effect vs. placebo	460 m	1.85 km	Retained
Slope	−17.47	−8.24	**47%**
Relief	−21.33	−11.80	**55%**
Depth	−7.55	−7.39	**98%**

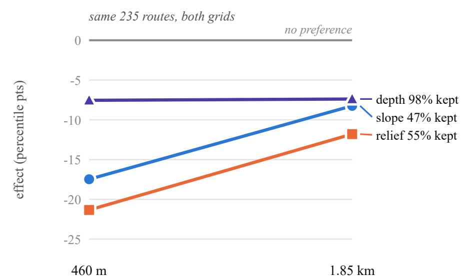

Fig. 4. The resolution gate: the same placebo measurement at both resolutions on the same routes. Depth survives almost intact; slope and relief lose about half. Had these collapsed toward zero, every long-haul result would be uninterpretable rather than negative.

Coarsening is legitimate only if the instrument can still see the effect. So the placebo measurement of Section IV-A was re-run at *both* resolutions on the same 235 routes before any long-haul number was quoted (Table II, Fig. 4). The sign is preserved throughout and 47%, 55% and 98% of the slope, relief and depth effects survive. Without this, a long-haul null would be indistinguishable from an instrument too blunt to see the effect.

E. Statistics

All significance is by permutation or sign test on per-route values; no distributional assumption is made about these heavily skewed quantities. Percentiles for the zero-inflated fishing surface use midranks, so a cable with both its controls in empty water scores 50 rather than 0.

V. RESULTS

A. Cables do not go straight

If cables essentially went straight there would be nothing to explain. Median sinuosity is 1.098 and median maximum lateral departure 17.5 km. However, 218 of 353 routes have geodesics that cross land: those cables *had* to bend, and "cables avoid continents" is true, trivial, and not a finding. Restricting to the 135 routes whose straight line was navigable open water, median sinuosity is 1.043 and median maximum departure 9.8 km.

On that informative subset, four of five terrain measures show nothing. The one signal is in the steep tail: 90th-percentile slope is lower on the observed route for 65% of

TABLE III
LOCAL TERRAIN PREFERENCE AGAINST THE PLACEBO-DISPLACED BASELINE (5 KM TRANSECTS, $n = 342$ ROUTES). A PERCENTILE BELOW THE PLACEBO MEANS THE CABLE SITS ON FLATTER, LESS RUGGED, OR SHALLOWER GROUND THAN THE WATER BESIDE IT. THE PLACEBO, NOT 50, IS THE ZERO POINT.

Metric	Real	Placebo	Effect	p
Slope	48.56	50.20	**−1.65**	$< 10^{-4}$
Roughness	48.77	50.30	**−1.54**	$< 10^{-4}$
Depth	50.71	50.29	**+0.42**	$< 10^{-4}$

routes ($p = 0.0005$). Depth runs the *opposite* way, with 61% of observed routes deeper than their straight line—consistent with following flat-bottomed troughs rather than minimising depth.

B. Local terrain preference is real and small

Table III gives the placebo-corrected result on $n = 342$ routes. Cables sit on measurably flatter and less rugged seabed than control lines drawn beside them—slope -1.65 and roughness -1.54 percentile points—and slightly *deeper*.

The control changed the conclusion. Raw, the depth effect read $+0.71$ against a null of 50 and looked substantial; baselined it is +0.42—most of it was the instrument, not the cables. Across all twelve placebo combinations the median percentiles sit at 50.0–50.6, eleven of twelve are non-significant, and the placebo *never* drops below 50 on slope or roughness. Effects also strengthen monotonically with transect width (raw slope 49.45 / 48.56 / 48.13 at 2 / 5 / 10 km), which is the gradient a real mechanism should show, since a wider transect offers more genuinely different ground to have chosen instead.

C. The dose–response that failed

If the mechanism is real it should appear where route choice is possible: over featureless shelf every position for tens of kilometres is equally flat, while over rugged ground there are real alternatives. Binning by terrain ruggedness gives a $3.5\times$ ratio between the flattest and ruggedest quartiles, in the predicted direction.

It does not hold. The series is not monotonic, and quoting a ratio from the two extreme bins of a noisy four-bin series is a number chosen after seeing it. Testing the trend across all 343 routes gives Spearman $\rho = -0.174$ ($p = 0.002$)—real but weak. More seriously, it is confounded with data source: the corpus is seven agencies whose survey fidelity differs by $30\times$ (Table I) and which survey different seas. Within each source, holding fidelity and cartographic convention constant, *no agency reaches significance* (ρ from -0.080 to -0.226, all $p > 0.05$). All four are negative, so the direction is consistent, but the gradient is largely between-source rather than terrain. The hypothesis is not supported.

D. Route prediction, regional

Table IV reports the headline numbers and Table V the paired tests that are actually sound.

TABLE IV
REGIONAL ROUTE PREDICTION, 40–500 KM AT 460 M, ON $n = 217$ HELD-OUT ROUTES. ERROR IS THE MEAN DISTANCE FROM THE CABLE ACTUALLY LAID, GIVEN ONLY THE TWO ENDPOINTS. BOLD BEATS THE GREAT CIRCLE. EVERY TERRAIN-AWARE ROW CARRIES THE 7.85 KM DISCRETISATION HANDICAP OF TABLE V'S FOOTNOTE; THE GREAT CIRCLE DOES NOT.

Method	Median (km)	Mean (km)	p90 (km)	vs. GC
Corridor + terrain	**6.4**	9.8	21.0	**−9.9%**
Corridor only	**6.9**	10.8	25.1	**−3.1%**
Great circle *(no bathymetry)*	7.1	11.2	26.3	+0.0%
Terrain: all terms, fitted	7.1	11.0	24.6	+0.2%
Terrain: slope only	7.3	10.7	22.6	+3.7%
Terrain: all terms, hand-set	7.9	11.2	24.9	+11.7%
Shortest sea path *(terrain ignored)*	9.7	13.6	30.3	+37.2%

TABLE V
PAIRED PER-ROUTE COMPARISONS, REGIONAL BAND. SIGN TEST ON THE SHARE OF ROUTES WHERE THE FIRST METHOD LANDS CLOSER TO THE AS-LAID CABLE. THESE COMPARISONS ARE THE SOUND ONES: BOTH MEMBERS SHARE THE GRID, THE SEARCH AND THE ENDPOINTS, SO THE DISCRETISATION HANDICAP CANCELS.

Comparison	n	Better on	p
Fitted terrain vs. shortest sea path	217	**62%**	7×10^{-4}
Corridor vs. shortest sea path	196	**66%**	$< 10^{-4}$
Corridor vs. *fitted* terrain	217	55%	0.135
Corridor vs. hand-set terrain	217	**60%**	**0.0028**
Adding terrain on top of corridor	217	53%	0.415

Discretisation control: a zero-weight grid search reproducing a known great circle still lands a median 7.85 km from it ($n = 51$ open-water pairs), which exceeds the 2.63 km gap the raw table attributes to bathymetry.

Read naively, Table IV says every bathymetry-aware method is worse than a straight line. *That reading is wrong*, and the discretisation control of Section IV-C is why: the 7.85 km grid handicap exceeds the whole geodesic–sea-path gap. The comparison that survives is between two grid searches, where the handicap cancels:

- **Terrain preference genuinely improves route prediction.** Median error falls from 9.7 km to 7.1 km—a 26% reduction—on 62% of the 217 held-out routes ($p = 0.0007$), across sources the weights were never fitted on. Slope alone reaches 7.3 km; the hand-set combination of all three terms is worse at 7.9 km.
- **Fitting the weights does not beat guessing them** (55%, $p = 0.14$ against fitted terrain; the fitted-versus-hand-set contrast is 52%, $p = 0.63$). Whatever the terrain signal is, it is coarse enough that a sensible hand-set weighting captures it—a direct and unflattering finding about learned cost surfaces for this problem, and one that bears on [6].

Fitted weights select slope in four of five folds, roughness in one and depth in three. Slope is the term that transfers, and it is the same term the placebo test found the strongest effect on, reached by a completely different method.

TABLE VI
DISTANCE TO THE NEAREST OTHER CABLE, REAL ROUTE VS. ITS PLACEBO-DISPLACED COPY, UNDER PROGRESSIVELY STRICTER CONTROLS. THE FIRST THREE ROWS EXCLUDE CANDIDATES BY IDENTITY; THE REST REMOVE THE LANDFALL GEOMETRY THAT IDENTITY EXCLUSION CANNOT TOUCH. THE BOLD ROW IS THE FIGURE THIS PAPER CLAIMS.

Control	n	Real (km)	Placebo (km)	Closer on	p
Any other route	336	2.1	9.8	90%	$< 10^{-4}$
Excl. same named cable	336	2.2	9.9	90%	$< 10^{-4}$
Excl. same agency	301	7.1	22.2	83%	$< 10^{-4}$
+ trim 10 km	227	9.3	22.7	79%	$< 10^{-4}$
+ trim 30 km	183	11.6	23.6	78%	$< 10^{-4}$
+ trim 50 km	158	11.5	22.3	79%	$< 10^{-4}$
+ trim 30, excl. shared landing	**182**	**19.0**	**26.6**	**74%**	$< 10^{-4}$

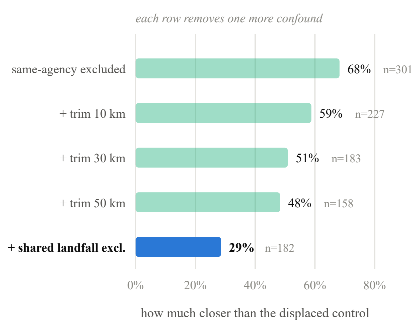

Fig. 5. The corridor effect as each confound is removed. Identity exclusion alone reports 68%; trimming the landfall approaches and then excluding cables that share a named landing point brings it to 29%. The first row is the figure a paper that stopped at identity exclusion would have published.

E. If not terrain, then what?

The most plausible alternative is not exotic: cable projects reuse corridors. Existing routes carry survey data, established permits, known burial conditions and proven landing approaches, all expensive to obtain afresh.

Using the identical placebo instrument, we measure the median distance from each route to the nearest *other* cable, then displace and measure again (Table VI). Three exclusion strengths guard the obvious confounds: a neighbouring segment of the same physical cable would make the result true by construction, and a single agency densely surveying one busy corridor says nothing about operators reusing *each other's* routes. The effect survives all three: even when a route may only be compared against cables published by a different country's hydrographic office, it sits 68% closer than the same line displaced sideways in the same water.

But identity exclusion is not enough, and this is where the uncontrolled figure fails. Cables converge at landing points because they have to. Two systems making landfall on the same beach are close there for reasons unrelated to corridor reuse, and a different operator's cable landing beside mine passes all three exclusions. Because displacement is applied per-vertex, it moves the real route's endpoints out of those convergence zones, so the real line samples crowded landfall water while its placebo samples emptier water beside it.

Two further controls are therefore applied, with the placebo receiving identical treatment. *Trimming the approaches* decays the effect and then *plateaus* near 50% (Table VI); pure landfall geometry would keep falling toward zero once the trim exceeded a convergence zone's width. *Excluding cables that land where I land*, using published landing points as an independent source, is stable across the association radius (31%/29%/26% at 5/10/20 km), which is what a real effect looks like and a threshold artefact does not.

So the honest figure is 29%, not 68% (Fig. 5). Roughly half the uncontrolled effect was the landfall approaches and a further part was cables genuinely sharing a landfall. What remains is corridor reuse in open water, on 74% of the 182 routes that survive both controls, at $p < 10^{-4}$—still several times the terrain effect, and requiring no survey data at all.

F. Corridor reuse as a competing router

Turning the observation into a model, we add a fourth cost term: distance to the nearest existing cable, zero at a cable and saturating 25 km away. Everything else is unchanged. The corridor term is fed a same-agency exclusion, because a router that can see the route it is predicting traces it and scores perfectly while measuring nothing.

Corridor following is the first and only method to beat the great circle (Table IV: 6.9 km alone, 6.4 km with terrain, against 7.1 km), and it does so while carrying the same 7.85 km handicap that made that comparison unfair for terrain. It beats the terrain-free baseline decisively (66%, $p < 10^{-4}$) and hand-set terrain (60%, $p = 0.003$).

Two cautions belong with that. It is *not* statistically distinguishable from *fitted* terrain at this sample size (55%, $p = 0.14$). And although corridor + terrain has the best median, adding terrain wins on only 53% of routes ($p = 0.42$): terrain helps markedly on some and hurts on others, so the median moves while the win rate sits at chance. Reporting "corridor + terrain is best" on the median alone would overstate what the paired test supports.

G. Does any of this hold for long-haul cables?

The strongest objection to the above is that it caps routes at 500 km and sits inside a regional bathymetry envelope, while the literature it argues with is about trans-oceanic systems. Testing on one regime and generalising to the other is a reason to reject, not a limitation to note in passing.

Table VII extends the experiment to every band up to 6,400 km on the 1.85 km grid, whose sensitivity cost is quantified in Section IV-D; Fig. 6 plots the same result relative to a great circle, and Table VIII gives the paired tests.

A reviewer's natural assumption is that corridor reuse is a shallow-water, landing-point artefact that will not survive in deep ocean. **The data says the reverse.** Corridor following is weakest on the shelf and dominant trans-oceanically, beating hand-set terrain on 15/15 routes above 2,000 km ($p = 3{\times}10^{-4}$) and the shortest sea path on 86% of 1,000–2,000 km routes.

TABLE VII
MEDIAN PREDICTION ERROR BY ROUTE LENGTH, ALL BANDS ON THE SAME 1.85 KM GRID. BOLD IS THE BEST METHOD IN THAT BAND. CORRIDOR FOLLOWING IMPROVES MONOTONICALLY WITH LENGTH; EVERY TERRAIN VARIANT DOES NOT.

Band	n	GC	Sea path	Terrain	Corridor	Cor+ter
			median error (km)			
40–500 km	253	7.1	9.8	8.1	6.5	**5.8**
500–1,000 km	47	43.1	51.6	**27.8**	37.6	29.8
1,000–2,000 km	28	149.4	86.1	84.3	52.7	**36.9**
2,000+ km	15	268.3	248.0	178.6	**55.6**	97.4

TABLE VIII
PAIRED PER-ROUTE COMPARISONS WITHIN EACH LENGTH BAND. THE NATURAL OBJECTION TO A REGIONAL CORPUS PREDICTS CORRIDOR FOLLOWING WEAKENING WITH DISTANCE; IT STRENGTHENS, AND ABOVE 2,000 KM ADDING TERRAIN ON TOP OF IT REVERSES SIGN.

Band	Comparison	n	Better on	p
40–500 km	Corridor vs. terrain	253	**62%**	3×10^{-4}
	Corridor vs. sea path	206	**71%**	$< 10^{-4}$
	Terrain added to corridor	232	51%	0.844
500–1,000 km	Corridor vs. terrain	47	49%	1.000
	Corridor vs. sea path	43	**74%**	**0.0023**
	Terrain added to corridor	47	62%	0.145
1,000–2,000 km	Corridor vs. terrain	28	68%	0.089
	Corridor vs. sea path	22	**86%**	**0.0014**
	Terrain added to corridor	28	46%	0.850
2,000+ km	Corridor vs. terrain	15	**100%**	3×10^{-4}
	Corridor vs. sea path	14	**93%**	**0.0033**
	Terrain added to corridor	15	27%	0.121

Above 2,000 km, adding terrain on top of corridor actively *hurts* (27%, $p = 0.12$). The regime the method is marketed for is the regime where it does worst against a bathymetry-free alternative.

The regional result also replicates at the coarser resolution—corridor 6.9→6.5 km, corridor + terrain 6.4→5.8 km, ordering unchanged—so the two grids can be set side by side rather than silently mixed.

H. Is the corridor result an artefact of a thin candidate set?

It could have been. Corridor distance above is measured against the 412-route corpus under same-agency exclusion, which for a transatlantic route leaves a handful of European candidates. Measured: the median distance to the nearest corpus candidate is 7.9 km at 40–500 km but 98.1 km above 2,000 km—beyond the 25 km saturation scale, so the cost term is flat across most of the route. The study's strongest result rested on its sparsest candidate set.

We therefore replace the candidate set entirely with 724 TeleGeography systems [16], which are dense in every band (5.1–9.2 km). Two problems had to be solved. Its geometry is schematic, so paths are densified to 2 km before indexing—a more *representative* instrument and a *noisier* one. And the subject cable is in the candidate set with no agency structure to exclude on. A fixed exclusion radius does not work here, and finding that out was useful: *no* corpus route has a TeleGeography system within 5 km, so a radius tight enough to mean "this is the same cable" fires on nothing, while one

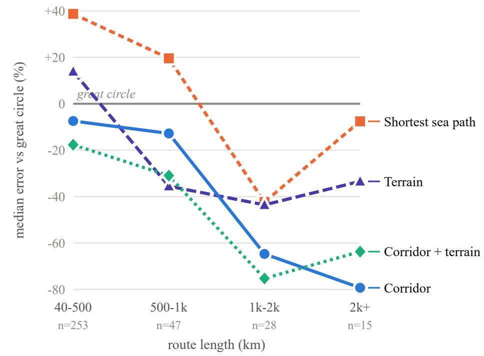

Fig. 6. Median prediction error relative to a great circle; below the line beats a straight line. Corridor following improves steadily with route length while every terrain variant flattens out—the opposite of what an objection to a regional-only corpus would predict.

loose enough to fire discards genuine corridor-mates. Systems are therefore ranked by how closely they shadow the subject and the top k dropped.

Table IX gives the result. **Long-haul survives**: an independent candidate set, an unrelated exclusion mechanism and the three closest-tracking systems removed, and corridor following still beats a great circle on 75% of 1,000–2,000 km routes and 15/15 above 2,000 km. **Regionally it fails.** The reason is measurable rather than mysterious, and it is the subject of the next section.

I. The error-ratio rule

Across bands, the ratio of the neighbour geometry's error to the great circle's error is 1.76 regionally (where corridor following hurts) and 0.17–0.18 long-haul (where it helps). A corridor known only to ±12 km cannot improve a 7 km prediction. This also explains why the surveyed corpus beats TeleGeography at 2,000+ km despite near-saturation: a handful of surveyed transatlantic routes is worth more than many schematic ones.

Three band-level points will fit almost any story, and length is confounded with sea, agency, depth and corridor density. So the ratio was computed for each route individually (Table X, Fig. 7). It is monotonic across every bin, with Spearman $\rho = 0.656$ on 348 routes ($z = 12.2$). Cutting at a ratio of 0.3, 86% of routes below it are improved by corridor following against 33% above it.

The rule holds *within* every length band, which makes it a statement about the mechanism rather than about length, and it is the most directly useful result here for anyone building such a router because it says in advance when the method will fail. It is descriptive, not causal: the ratio and the gain share the great-circle error as a denominator, so a route that is simply hard to predict moves both. The binned pattern is the load-bearing part; the correlation coefficient inherits that shared term and should not be read as an effect size.

TABLE IX

REPLACING THE CANDIDATE SET. EACH CELL IS MEDIAN ERROR (KM) / SHARE OF ROUTES BEATING A GREAT CIRCLE, FOR THE EMODNET CORPUS UNDER SAME-AGENCY EXCLUSION AND FOR 724 TELEGEOGRAPHY SYSTEMS WITH THE k CLOSEST-TRACKING SYSTEMS REMOVED. BOLD IS SIGNIFICANTLY BETTER THAN A GREAT CIRCLE, <u>UNDERLINED</u> SIGNIFICANTLY WORSE. LONG-HAUL SURVIVES AN INDEPENDENT CANDIDATE SET AND AN UNRELATED EXCLUSION MECHANISM; THE REGIONAL BAND DOES NOT.

Band	n	Corpus	TG k=0	TG k=1	TG k=3
40–500 km	259	6.4 / **57%**	7.3 / 52%	11.0 / <u>36%</u>	12.3 / <u>32%</u>
500–1,000 km	47	37.6 / 57%	28.5 / 64%	39.0 / 53%	39.1 / 55%
1,000–2,000 km	28	52.7 / **82%**	36.2 / **79%**	63.0 / **79%**	79.7 / **75%**
2,000+ km	15	55.6 / **100%**	130.1 / **100%**	149.6 / **100%**	155.5 / **100%**

TABLE X

THE ERROR-RATIO RULE, TESTED PER ROUTE RATHER THAN PER BAND ($n = 348$). THE RATIO IS THE NEIGHBOUR GEOMETRY'S ERROR OVER THE GREAT CIRCLE'S ERROR FOR THAT SAME ROUTE. MONOTONIC ACROSS EVERY BIN, SPEARMAN $\rho = 0.656$ ($z = 12.2$).

Ratio range	n	Median ratio	Routes helped	Median gain
0.04–0.26	58	0.15	**90%**	−52.7%
0.26–0.49	58	0.36	67%	−22.9%
0.51–0.93	58	0.69	41%	+22.9%
0.96–1.58	58	1.25	31%	+44.3%
1.59–3.63	58	2.46	29%	+73.6%
3.73–92.88	58	6.73	**7%**	+352.1%

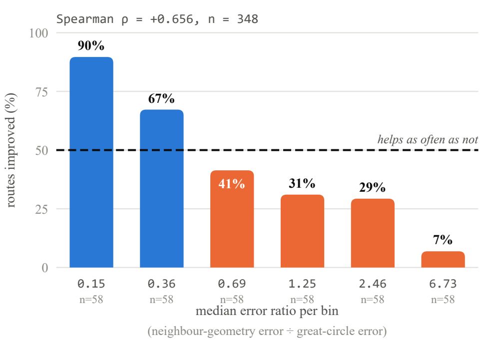

Fig. 7. When corridor following helps, per route rather than per band. Above the dashed line the method helps more often than not.

J. Do cables avoid fishing grounds?

External aggression—overwhelmingly fishing gear and anchors, in water shallower than about 200 m—accounts for roughly three quarters of recorded cable faults [22]. If cables are not following terrain, the natural next hypothesis is that they are avoiding the hazard that actually breaks them.

There is no fishing-avoidance signal (Table XI, Fig. 8). What effect exists points the wrong way—cables in slightly *heavier* fishing—and it vanishes once 50 km of each end is removed. It is landfall geometry: coastal water is busy, and a perpendicular control near shore moves offshore into quieter water. This is the same confound that took the corridor effect from 68% to 29%. Depth stratification agrees: at 5 km displacement the effect is 54.6 on the shelf, 51.6 on the slope and 49.9 in deep water—confined to exactly the zone where a coastal artefact would live, and absent where trawling avoidance would have to show up if it were real.

One sub-design failed and is reported as untested rather than rejected. Cable protection zones restrict fishing near cables, so light fishing beside a cable could be a *consequence* of the cable. The displacement sweep was designed to discriminate this—suppression predicts a step as the control leaves the zone, route choice a smooth rise—but at ∼1.7 km native

TABLE XI

FISHING EFFORT: MEAN PERCENTILE MINUS 50, BY CONTROL DISPLACEMENT AND BY HOW MUCH OF EACH ROUTE END IS DISCARDED. POSITIVE MEANS THE CABLE SITS IN HEAVIER FISHING THAN THE WATER BESIDE IT. * $p < 0.05$ ACROSS ROUTES. THE EFFECT IS CONFINED TO THE UNTRIMMED COLUMNS AND VANISHES AT A 50 KM TRIM, WHICH LOCATES IT AT THE LANDFALLS.

Displacement	No trim	10 km	25 km	50 km
3 km	**+0.90***	**+1.34***	**+1.30***	+0.24
5 km	**+2.05***	**+1.34***	**+1.47***	+0.15
10 km	+0.92	+0.62	+0.65	−0.28
20 km	+0.86	+0.11	−0.43	−0.50
50 km	+1.21	+0.53	−0.09	−0.40
routes	*230*	*203*	*170*	*141*

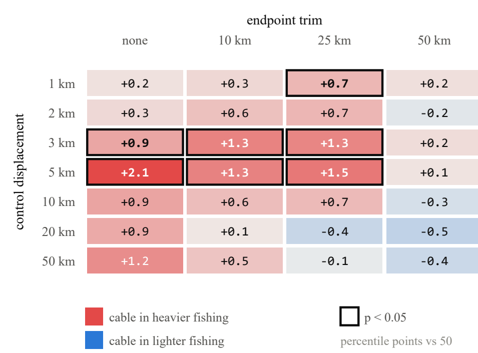

Fig. 8. Fishing effect against endpoint trim and control displacement. Outlined cells are significant across routes. The effect is confined to the untrimmed and lightly trimmed columns and disappears at a 50 km trim, which locates it at the landfalls rather than along the route. The 1 and 2 km rows are shown for completeness but are not interpretable and are omitted from Table XI: at ∼1.7 km native resolution a control displaced that far usually lands in the same cell and scores 50 by construction.

resolution a control displaced 1–2 km usually lands in the same cell and scores 50 by construction. A typical protection zone is narrower than one native cell, and no resampling fixes that.

VI. Threats to Validity

A. Measurement floor

The flattest ruggedness quartile has local relief of 0.0–1.5 m, the same order as the downsampling error measured on those same shelf tiles (0.46–1.28 m RMS). Its near-null effect is partly an instrument limit, not demonstrated absence of preference.

One source contradicts the main effect: DE BSH-CONTIS shows cables on *steeper* ground (Δslope $+1.57$). It is also the only source whose median terrain relief (1.0 m) falls below the grid's own downsampling error (1.28 m RMS on shelf tiles). Measured across all sources, the relief-to-noise ratio is 0.78 for BSH and 1.56–21.5 elsewhere, and every source above the floor agrees with the main finding, the one sitting marginally at the boundary agreeing most weakly. The contradiction tracks the measurement floor, not the agency. We report this as a *scope condition*—the method needs relief above the bathymetry's noise—rather than as an unexplained result.

B. Scope and structure

Geographic coverage. 75% of corpus length is NW Europe and the Mediterranean. The long-haul extension does not fix this; it *sharpens* it. The 2,000+ km band is 13 of 15 routes from one agency and reduces in practice to two corridors, transatlantic and Marseille–Levant, at $n = 15$. Broadening it needs as-laid geometry from non-European hydrographic offices, which is a data-acquisition project rather than an analysis one.

Fold structure collapses above 1,000 km. Fewer than two sources reach the eight-route minimum, so leave-one-source-out cannot run there. The fixed-weight methods carry those results and need no fitting, but the spatially blocked design that protects Section V-D does not protect Section V-G; the fitted method is reported as unavailable rather than quietly fitted on the routes it is tested on.

Two resolutions are in play. The coarser grid retains 47% of the slope effect and 55% of relief (Table II), so long-haul terrain results carry roughly half the sensitivity. This biases *against* terrain in the comparison where terrain loses, which is the conservative direction, but it is not neutral and figures should not be quoted across the two.

The corridor bound. Bathymetry was loaded within $\pm 1^\circ$ ($\sim$111 km) of each route—generous for the median 171 km route, tight for the 49 routes over 1,000 km, whose plausible alternatives may lie outside it.

Regional k-sweep ambiguity. In the regional band the k-sweep of Section V-H cannot cleanly separate "removed the leak" from "removed the signal", because a schematic entry for the subject and for its genuine corridor-mates sit at similar distances. The long bands do not have this problem.

Temporal direction. The fishing surface is 2022 and most corpus cables were laid earlier. Nothing in Section V-J can establish that planners avoided fishing grounds; at most it establishes that cables and fishing effort are not spatially separated today.

Unmeasured criteria. Bathymetry, existing infrastructure and fishing intensity are measured here. Sediment type, anchoring zones, protected areas and jurisdiction are not.

VII. Discussion

A. What the numbers say to a tool builder

Cables do prefer flatter, less rugged seabed. The effect is real, controlled, and consistent in sign across five of six independent national sources—and it is small, and it does not scale with terrain difficulty in any way that survives controlling for who published the data.

The most useful reading is the negative one: **bathymetry exerts a weak influence on local cable position**, and the dominant factors are elsewhere. Of the four candidate drivers weighed here—terrain, existing infrastructure, fishing, and everything unmeasured—the second dominates, the first is small, the third is absent, and a residual error of 6.4 km against a 7.85 km grid handicap says the fourth is still doing most of the work: landing-point contracts, permitting, seasonal vessel availability and commercial relationships.

This is a direct challenge to the premise of terrain-driven least-cost-path cable routing. It does not make such tools useless—a route ignoring terrain would be worse—but their output should be presented as one input to a decision rather than as an optimum. Two consequences are specific enough to act on:

Do not over-engineer the cost surface. Fitting weights does not beat guessing them. Effort spent learning a terrain cost function is better spent elsewhere, which bears directly on the approach of [6].

Add the corridor term, and check the ratio first. It is the strongest single predictor found here, it needs no bathymetry, and Section V-I says in advance whether it will help on a given route.

B. The corridor result has a policy counterpart

Corridor reuse is not a neutral finding. Regulators and industry bodies have examined *spatial separation* of cables as a resilience measure [23], on the reasoning that co-located systems share fate under a single anchor drag or seabed event. Our measurement says co-location is not merely permitted but is a strong empirical regularity in where cables actually go, and that it grows stronger with distance. Whether that reflects rational reuse of survey and permitting effort or an accumulating concentration risk is outside what this dataset can answer, but the quantification belongs in that debate.

C. Reproducibility

Every number in this paper is produced by a script in the public repository, writes its result to a cache, and is read from that cache by the table and figure generators—so a figure or table disagreeing with the text means one of them is stale rather than that a number was mistyped. The pipeline requires no API key and no account; the only prerequisite is Node. Total acquisition is about 615 MB, most of it bathymetry, and after the cache is built every test except the weight fit re-runs in under a minute combined.

The two load-bearing results are the cheapest to check. The corridor effect is measured purely on route geometry and touches no bathymetry, so it needs the corpus build alone—a couple of minutes, no tiles. The discretisation control, which reversed the sign of the headline in Section V-D, additionally needs the regional bathymetry.

VIII. Conclusion

We tested the premise underlying least-cost-path submarine cable routing against 412 as-laid routes and found it weak. Terrain preference is real, small, does not strengthen with terrain difficulty once the publishing agency is controlled for, and is captured as well by hand-set weights as by fitted ones. A bathymetry-free alternative—proximity to existing cables—outperforms it, survives four independent controls and a complete replacement of its candidate set, strengthens with route length, and carries a measured rule that predicts when it will fail. Fishing effort, the most plausible remaining explanation, shows no signal at all.

Three controls made those findings possible, and each reversed or substantially changed a headline: a placebo-displaced baseline, a discretisation penalty, and a resolution gate. Their common lesson is that an evaluation of route-choice models needs a stated zero point for its instrument, and neither the nominal null, nor the analytic baseline, nor the finer grid the instrument was calibrated on is automatically that zero point. Each has to be measured.

The obvious extension is geographic. The corpus is Europe-weighted because that is where as-laid geometry is published; testing whether corridor reuse holds in the Pacific, where cable systems are younger and corridors sparser, is a data-acquisition problem and the single most valuable thing that could be done next.

Acknowledgment

This work uses open data published by EMODnet Human Activities, EMODnet Bathymetry, and the NOAA National Centers for Environmental Information.

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